

UNIVERSITY OF KARACHI
UBIT - Department of Computer Science

Ultimate DLD Master Bank

Digital Logic Design (SE-455)

Exclusive MCQ Practice Sheet

This sheet exclusively contains all practice MCQs and rigorously excludes theory.

270+ High-Yield, In-Depth Solved MCQs

*Covers Theory, Conversions, Gates, Adders,
FFs, Timers, Hardware, VHDL & Memory*

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Chapter 1: Number Systems, Conversions and Codes

1. A Continuous or Analog function is mathematically represented by which type of graph?
 - A. A discrete binary tree.
 - B. **A sinusoidal graph that changes continuously at every point.** ✓
 - C. A straight constant line.
 - D. A Karnaugh Map.
2. Digital hardware exclusively understands which type of function?
 - A. **Discrete / Digital functions (existing strictly in 1 or 0 states).** ✓
 - B. Continuous / Analog functions.
 - C. High-level language functions.
 - D. Hexadecimal functions.
3. The mathematical expression $3 \times 10^1 + 4 \times 10^0$ evaluates to:
 - A. 0.34
 - B. 3.4
 - C. **34** ✓
 - D. 340
4. The radix (or base) of the binary number system is:
 - A. **2** ✓
 - B. 8
 - C. 10
 - D. 16
5. The Octal number system is a weighted system with how many distinct digits?
 - A. 2
 - B. **8 (Digits 0 to 7)** ✓
 - C. 10
 - D. 16
6. To rapidly convert a Binary number to an Octal number, you must group the binary bits into sets of:
 - A. 2 bits
 - B. **3 bits (Starting from LSB)** ✓
 - C. 4 bits
 - D. 8 bits
7. To rapidly convert a Binary number to a Hexadecimal number, you must group the binary bits into sets of:

- A. 2 bits
 - B. 3 bits
 - C. **4 bits (Starting from LSB) ✓**
 - D. 8 bits
8. What is the maximum unsigned decimal value you can represent using exactly 4 bits?
- A. 8
 - B. 10
 - C. **15 (Since $2^4 - 1 = 15$) ✓**
 - D. 16
9. What is the highest decimal digit you can legally represent in a standard BCD (8421) code?
- A. 7
 - B. **9 ✓**
 - C. 10
 - D. 15
10. The decimal equivalent of the binary number 1000 is:
- A. 2
 - B. 4
 - C. 6
 - D. **8 ✓**
11. The decimal number 21 is mathematically equivalent to which binary number?
- A. **10101 ✓**
 - B. 10001
 - C. 10000
 - D. 11111
12. The binary number 11011101 is equal to the decimal number:
- A. 121
 - B. **221 ✓**
 - C. 441
 - D. 256
13. The decimal number 250 is equivalent to the binary number:
- A. 11110110
 - B. **11111010 ✓**
 - C. 11111100
 - D. 11111000
14. The sum of binary numbers 111 + 1111 strictly equals:

- A. 0000
 - B. 2222
 - C. **10110** ✓
 - D. 11110
15. The difference of binary numbers $1000 - 100$ equals:
- A. **100** ✓
 - B. 101
 - C. 110
 - D. 111
16. What is the range of decimal values that can be represented by an 8-bit signed integer using 2's complement?
- A. 0 to 255
 - B. -127 to +127
 - C. **-128 to +127** ✓
 - D. -255 to +255
17. The Hexadecimal number system consists of 16 characters. How many of these are alphabetic characters?
- A. 4
 - B. **6 (A, B, C, D, E, F)** ✓
 - C. 8
 - D. 10
18. If you add $9 + 1$ strictly in Hexadecimal logic, the correct mathematical result is:
- A. 10
 - B. **A** ✓
 - C. B
 - D. 11
19. How do you physically find the 1's complement of a binary number?
- A. By adding 1 to the LSB.
 - B. **By inverting all the bits (changing 1s to 0s and 0s to 1s).** ✓
 - C. By shifting the bits to the left.
 - D. By multiplying by 2.
20. What is the 1's complement of the binary number 11110000?
- A. 11111111
 - B. 11111110
 - C. **00001111** ✓
 - D. 10000001

21. What is the 2's complement of the binary number 11001100?
- A. 00110011
 - B. **00110100** ✓
 - C. 00110101
 - D. 00110110
22. In a signed binary number representation, what does the left-most bit (MSB) formally represent?
- A. The parity bit.
 - B. The total magnitude.
 - C. **The sign bit (0 for positive, 1 for negative).** ✓
 - D. The least significant bit.
23. A single-precision floating-point binary number (IEEE 754) has a total of:
- A. 8 bits
 - B. 16 bits
 - C. 24 bits
 - D. **32 bits** ✓
24. A single-bit error occurring during data transmission can be easily detected using:
- A. A 2's complement code.
 - B. **An odd or even Parity bit.** ✓
 - C. A multiplexer.
 - D. An ASCII converter.
25. Which of the following codes exhibits an EVEN parity error (assuming it's supposed to be even)?
- A. 1010011
 - B. **1101000 (It has an odd number of 1s, breaking the even parity rule)** ✓
 - C. 1001000
 - D. 1110111
26. Which specific error-detecting code is primarily used in digital networks and storage devices, standing for Cyclic Redundancy Check?
- A. BCD
 - B. **CRC** ✓
 - C. ASCII
 - D. Gray Code
27. In a Cyclic Redundancy Check (CRC), the absolute absence of errors is indicated when:
- A. Remainder = generator code
 - B. **Remainder = 0** ✓

- C. Remainder = 1
 - D. Quotient = 0
28. The modulo-2 sum of binary numbers 11 and 10 is effectively an XOR operation. What is the result?
- A. 100
 - B. **01 ✓**
 - C. 101
 - D. 00
29. What does BCD specifically stand for in digital electronics?
- A. Binary Converted Data
 - B. **Binary Coded Decimal ✓**
 - C. Boolean Coded Decimal
 - D. Binary Continuous Data
30. Which of the following is considered an "Unweighted" code?
- A. Binary
 - B. BCD (8421)
 - C. Octal
 - D. **Excess-3 ✓**
31. Which binary code is uniquely characterized by the fact that only a single bit changes between any two successive numbers?
- A. BCD
 - B. Excess-3
 - C. ASCII
 - D. **Gray Code (Reflected Binary Code) ✓**
32. The standard, worldwide 7-bit encoding method used to convert human-readable alphabets into binary form for computers is:
- A. **ASCII ✓**
 - B. Modulo-2
 - C. EBCDIC
 - D. Gray Code
33. What does EBCDIC stand for?
- A. Electronic Binary Coded Data Interchange Code
 - B. **Extended Binary Coded Decimal Interchange Code ✓**
 - C. Enhanced Binary Continuous Digital Interface Code
 - D. Error-free Binary Coded Decimal Information Code
34. The command "STOP" in ASCII encoding is represented by a sequence of bits. Knowing 'S' is 1010011 and 'T' is 1010100, which sequence logically matches?

- A. **1010011 1010100 1001111 1010000** ✓
- B. 1001010 1101101 1001110 1010001
- C. 1010011 1010010 0111101 0000000
- D. 0100011 1001100 1010011 1001011
35. The binary number 10001101010001101111 can be written in hexadecimal as:
- A. AD467
- B. **8D46F (Grouped as 1000 1101 0100 0110 1111)** ✓
- C. 8C46F
- D. AE46F
36. The binary number for the Hexadecimal value $F7A9_{16}$ is:
- A. 1111011110101001
- B. **1111011110101001** ✓
- C. 1111111010110001
- D. 1111011010101001
37. The fractional binary number $(0.101)_2$ represents which decimal value?
- A. 0.500
- B. **0.625 (Since $1/2 + 0/4 + 1/8 = 0.625$)** ✓
- C. 0.125
- D. 0.750

Chapter 2: Boolean Algebra, Laws and Logic Gates

38. A variable is a symbol in Boolean algebra universally used to represent:
- A. Data only
 - B. A condition only
 - C. An action only
 - D. **Data, conditions, or actions (all of these).** ✓
39. The Boolean expression $A + B + C$ is structurally identified as:
- A. **A sum term** ✓
 - B. A literal term
 - C. An inverse term
 - D. A product term
40. The Boolean expression $\overline{ABCD}$ is structurally identified as:
- A. A sum term
 - B. A literal term
 - C. **An inverse term (or a negated product term)** ✓
 - D. A base term
41. The mathematical "domain" of the Boolean expression $A\overline{B}CD + A\overline{B} + \overline{C}D + B$ is:
- A. A and D only
 - B. B only
 - C. **A, B, C, and D** ✓
 - D. A, B, and C only
42. According to the commutative law of multiplication in Boolean algebra:
- A. $A(B + C) = AB + AC$
 - B. **$AB = BA$** ✓
 - C. $(AB)C = A(BC)$
 - D. $A \cdot 1 = A$
43. According to the associative law of addition in Boolean algebra:
- A. $A + B = B + A$
 - B. $A + (AB) = A$
 - C. **$(A + B) + C = A + (B + C)$** ✓
 - D. $A + A = A$
44. According to the distributive law in Boolean algebra:
- A. **$A(B + C) = AB + AC$** ✓

- B. $AB = BA$
C. $A(A + 1) = A$
D. $A + AB = A$
45. Which of the following rules strictly states that if one input of an AND gate is always 1, the output is exactly equal to the other input?
- A. $A + 1 = 1$
B. $A + A = A$
C. $A \cdot A = A$
D. $A \cdot 1 = A$ ✓
46. Which basic logic gate produces a HIGH (1) output if ANY of its inputs are HIGH?
- A. AND Gate
B. **OR Gate** ✓
C. NOT Gate
D. NAND Gate
47. Which universal logic gate produces a LOW (0) output ONLY when all of its inputs are HIGH (1)?
- A. AND Gate
B. OR Gate
C. **NAND Gate** ✓
D. NOR Gate
48. According to De Morgan's Theorem, the complement of a product $(A \cdot B)'$ is mathematically equal to:
- A. $A' \cdot B'$
B. $A' + B'$ ✓
C. $A + B$
D. $(A + B)'$
49. According to De Morgan's Theorem, the complement of a sum $(A + B)'$ simplifies mathematically to:
- A. $A' \cdot B'$ ✓
B. $A' + B'$
C. $A \cdot B$
D. $A' \cdot B$
50. Which two logic gates are universally known as "Universal Gates" because any digital circuit can be built entirely using only them?
- A. AND and OR
B. **NAND and NOR** ✓
C. XOR and XNOR

- D. NOT and AND
51. To implement a basic OR gate using ONLY NAND gates, what is the minimum number of NAND gates required?
- A. 1
 - B. 2
 - C. 3 ✓
 - D. 4
52. To implement a basic AND gate using ONLY NOR gates, what is the minimum number of NOR gates required?
- A. 1
 - B. 2
 - C. 3 ✓
 - D. 4
53. A NAND gate is logically equivalent to an OR gate with inverted inputs. This is officially known as a:
- A. Positive OR gate
 - B. Negative-OR (Bubbled OR) gate ✓
 - C. XOR gate
 - D. Buffer
54. According to Boolean Algebra (Idempotent Law), the expression $A + A$ is equal to:
- A. $2A$
 - B. 1
 - C. A ✓
 - D. 0
55. According to the Absorption Law in Boolean Algebra, $A + AB$ simplifies to:
- A. B
 - B. A ✓
 - C. $A + B$
 - D. 1
56. Which important Boolean rule simplifies the expression $A + A'B$ to just $A + B$?
- A. De Morgan's Law
 - B. Commutative Law
 - C. Rule 11 / Simplification Law ✓
 - D. Involution Law
57. According to Boolean Algebra rules, the expression $A + 1$ is always equal to:
- A. A

- B. 0
- C. **1** ✓
- D. A'

58. According to Boolean Algebra rules, the expression $A \cdot 0$ is always equal to:

- A. A
- B. **0** ✓
- C. 1
- D. A'

59. According to Boolean Algebra rules, the expression $A \cdot A'$ is always equal to:

- A. A
- B. **0** ✓
- C. 1
- D. A'

60. According to Boolean Algebra rules, the expression $A + A'$ is always equal to:

- A. A
- B. 0
- C. **1** ✓
- D. A'

61. The Involution Law (Double Negation) in Boolean algebra states that $(A')'$ is equal to:

- A. 0
- B. 1
- C. A'
- D. **A** ✓

62. Which logic gate is commonly known as the "Inequality Gate" because it outputs a 1 ONLY when its inputs are different from each other?

- A. **XOR (Exclusive-OR)** ✓
- B. XNOR
- C. NAND
- D. AND

63. The Boolean equation $A \oplus B$ (which represents A XOR B) is mathematically equivalent to:

- A. $A'B'$
- B. $AB + A'B'$
- C. **$A'B + AB'$** ✓
- D. $A + B$

64. Which logic gate outputs a 1 ONLY when its inputs are exactly the same (Equality Gate / Coincidence Gate)?

- A. XOR Gate
 - B. **XNOR Gate** ✓
 - C. NAND Gate
 - D. NOR Gate
65. The Boolean equation for an Exclusive-NOR (XNOR) function is mathematically expressed as:
- A. $A'B'$
 - B. $AB + A'B'$ ✓
 - C. $A'B + AB'$
 - D. $A + B$
66. Which basic logic gate can be universally utilized as a Controlled Inverter?
- A. AND Gate
 - B. OR Gate
 - C. **XOR Gate** ✓
 - D. Buffer
67. The maximum number of inputs a standard logic gate can successfully drive without signal degradation is known as its:
- A. Fan-in
 - B. **Fan-out** ✓
 - C. Propagation Delay
 - D. Duty Cycle
68. The time it takes for a change in the input of a logic gate to produce a corresponding change at its output is defined as:
- A. Fan-out
 - B. Hold Time
 - C. **Propagation Delay (t_p)** ✓
 - D. Setup Time
69. What does VHDL stand for in digital design?
- A. Visual Hardware Definition Language
 - B. Variable High-Level Design Language
 - C. **VHSIC Hardware Description Language** ✓
 - D. Virtual Hardware Description Logic
70. Using VHDL, a logic circuit's external view (inputs and outputs) is formally described in the:
- A. Architecture
 - B. **Entity** ✓
 - C. Component

D. Data Flow

71. Using VHDL, a logic circuit's internal behavior and structure are described in the:

- A. **Architecture** ✓
- B. Entity
- C. Component
- D. Port

72. In PLD (Programmable Logic Device) terminology, what does JTAG stand for?

- A. Joint Time Analysis Group
- B. Joint Test Array Group
- C. **Joint Test Action Group** ✓
- D. Java Top Array Group

73. An antifuse in programmable logic is typically formed using:

- A. Two insulators separated by a conductor
- B. **Two conductors separated by an insulator** ✓
- C. A transistor array
- D. Two diodes connected in series

Chapter 3: Combinational Logic and K-Maps

74. A Combinational Logic Circuit is explicitly defined as a circuit whose output depends ONLY on:
- A. The previous output state.
 - B. **The present inputs at that exact moment. ✓**
 - C. The clock pulse.
 - D. The memory registers.
75. What is the primary purpose of utilizing a Karnaugh Map (K-Map) in Digital Logic Design?
- A. To convert binary to hexadecimal.
 - B. **To visually simplify Boolean expressions and minimize the number of logic gates needed. ✓**
 - C. To increase the voltage of a circuit.
 - D. To act as a memory storage device.
76. How many total cells are explicitly required in a K-Map used to simplify an expression with exactly 3 variables?
- A. 4
 - B. **8 (Calculated as 2^3) ✓**
 - C. 16
 - D. 32
77. How many total cells are explicitly required in a K-Map used to simplify an expression with exactly 4 variables?
- A. 4
 - B. 8
 - C. **16 (Calculated as 2^4) ✓**
 - D. 32
78. How many total cells are required to construct a 5-variable K-map (usually represented as two 4-variable maps stacked)?
- A. 16
 - B. **32 (Calculated as 2^5) ✓**
 - C. 64
 - D. 128
79. Which of the following is a strict mathematical rule when grouping cells (1s) in a Karnaugh Map?
- A. You can group cells diagonally.
 - B. **Groups must contain a number of cells equal to a power of 2 (1, 2, 4, 8, etc.). ✓**

- C. Groups cannot overlap under any circumstances.
 - D. You must group 0s and 1s together.
80. In K-Map mapping, the concept where the extreme left column is considered adjacent to the extreme right column is known as:
- A. Overlapping
 - B. **Wrapping Around (Map Rolling) ✓**
 - C. Redundancy
 - D. Parity Checking
81. When grouping cells in a K-Map, what is your primary objective?
- A. To create as many small groups as possible.
 - B. To group all the 0s instead of 1s.
 - C. **To create the largest possible groups of 1s (e.g., octets over quads) to yield the simplest expression. ✓**
 - D. To avoid overlapping any groups.
82. In K-Map terminology, a grouping of 1s that forms a valid product term is called a:
- A. **Prime Implicant ✓**
 - B. Maxterm
 - C. Redundant Group
 - D. Floating state
83. If a specific '1' in a K-Map can ONLY be covered by exactly one unique group, that group is officially known as a/an:
- A. Redundant Implicant
 - B. **Essential Prime Implicant ✓**
 - C. Minterm
 - D. Overlapped Group
84. By grouping a Pair (exactly 2 cells) in a K-Map, how many variables are logically eliminated from that final Boolean product term?
- A. **1 variable ✓**
 - B. 2 variables
 - C. 3 variables
 - D. 0 variables
85. By grouping a Quad (exactly 4 cells) in a K-Map, how many variables are logically eliminated from that final Boolean product term?
- A. 1 variable
 - B. **2 variables ✓**
 - C. 3 variables
 - D. 0 variables

86. In a 4-variable K-Map, if you group the four extreme corners together, how many variables are eliminated?
- A. 1 variable
 - B. **2 variables ✓**
 - C. 3 variables
 - D. 4 variables
87. By grouping an Octet (exactly 8 cells) in a K-Map, how many variables are logically eliminated?
- A. 1 variable
 - B. 2 variables
 - C. **3 variables ✓**
 - D. 4 variables
88. A "Don't Care" condition (represented by X or d) in a K-Map can be treated as:
- A. Strictly a 0.
 - B. Strictly a 1.
 - C. **Either a 0 or a 1, whichever helps create the largest possible group. ✓**
 - D. An error state.
89. When a K-Map becomes too visually complex (handling 5 or more variables), which advanced tabular method is strictly utilized for minimization?
- A. De Morgan's Method
 - B. **Quine-McCluskey Method ✓**
 - C. Boolean Algebra Expansion
 - D. Dijkstra's Method
90. In Sum of Products (SOP) form, a standard product term that contains ALL the variables of the function is formally called a:
- A. Maxterm
 - B. **Minterm (Canonical Form) ✓**
 - C. Literal
 - D. Prime Implicant
91. Product of Sums (POS) expressions are typically derived from K-Maps by actively grouping which values?
- A. The 1s
 - B. **The 0s ✓**
 - C. The Don't Cares only
 - D. The MSB
92. A momentary, unwanted spike or pulse at the output of a combinational logic circuit due to uneven propagation delays in different gates is called a:

- A. Metastable state
 - B. **Static Hazard (or Glitch)** ✓
 - C. Clock Skew
 - D. Race Condition
93. An example of a strictly "Standard SOP" (Canonical) expression is:
- A. $A + B(C + D)$
 - B. $ABCD + AB + A$
 - C. $\overline{A}BC + A\overline{B}C + ABC$ (All variables appear in each term) ✓
 - D. $A + B + C$
94. The acronym AOI in digital logic stands for AND-OR-Invert, which is primarily used to directly implement:
- A. SOP expressions
 - B. **POS (Product of Sums) expressions** ✓
 - C. Only XOR functions
 - D. Shift registers
95. Which of the following is a classic example of a Product-of-Sums (POS) expression?
- A. $(A + B)(\overline{A} + B + \overline{C})$ ✓
 - B. $A(B + C) + A\overline{C}$
 - C. $A + B + BC$
 - D. $AB + CD$
96. In VHDL terminology, a "component" is best defined as:
- A. A simple wire connection.
 - B. **A predefined description of a logic function that can be used multiple times.** ✓
 - C. A method of saving files.
 - D. A type of clock edge.
97. In VHDL, bringing a component into the current design for use is technically called:
- A. A variable declaration
 - B. **Component instantiation** ✓
 - C. A structural delay
 - D. An entity tag

Chapter 4: Advanced Combinational Circuits (Adders, Sub, Encoders, MUX)

98. A Half Adder circuit is designed to add two single bits. Its 'Sum' output is mathematically generated by which logic gate?
- A. AND Gate
 - B. OR Gate
 - C. **XOR Gate ✓**
 - D. NAND Gate
99. A Half Adder circuit's 'Carry' output is mathematically generated by which logic gate?
- A. **AND Gate ✓**
 - B. OR Gate
 - C. XOR Gate
 - D. NAND Gate
100. To mathematically construct a Full Adder circuit from scratch, what minimum components are required?
- A. 1 Half Adder and 1 OR Gate
 - B. **2 Half Adders and 1 OR Gate ✓**
 - C. 3 Half Adders
 - D. 2 XOR Gates and 2 AND Gates
101. The standard Boolean equation for a Full Adder's "Carry-Out" (C_{out}) given inputs A, B, and C_{in} is:
- A. $A \oplus B \oplus C_{in}$
 - B. **$(A \cdot B) + (C_{in} \cdot (A \oplus B))$ ✓**
 - C. $A \cdot B \cdot C_{in}$
 - D. $A + B + C_{in}$
102. A 3-bit parallel adder is designed to add together:
- A. Three 2-bit binary numbers
 - B. **Two 3-bit binary numbers ✓**
 - C. Three bits at a time (serial)
 - D. Three bits in sequence
103. To practically expand a 2-bit parallel adder to a 4-bit parallel adder, what must be done?
- A. Use two 2-bit adders with no interconnections.
 - B. **Use two 2-bit adders with the carry output (C_{out}) of one connected to the carry input (C_{in}) of the other. ✓**
 - C. Connect sum outputs to inputs directly.

- D. Replace all XOR gates with AND gates.
104. The "Difference" output of a Half Subtractor (given inputs A and B) is generated by which equation?
- A. $A \oplus B$ ✓
 - B. $A \cdot B'$
 - C. $A' \cdot B$
 - D. $(A \oplus B)'$
105. The "Borrow-Out" output of a Half Subtractor (where A is minuend and B is subtrahend) is:
- A. $A \cdot B$
 - B. $A' \cdot B$ ✓
 - C. $A \cdot B'$
 - D. $A + B'$
106. The "Difference" output of a Full Subtractor (given A, B, and Bin) is:
- A. $A \cdot B \cdot Bin$
 - B. $A \oplus B \oplus Bin$ ✓
 - C. $(A \oplus B)' \cdot Bin$
 - D. $A + B + Bin$
107. The "Borrow-Out" equation of a Full Subtractor is mathematically equivalent to:
- A. $A'B + Bin \cdot (A \oplus B)$
 - B. $A'B + Bin \cdot (A \oplus B)'$ ✓
 - C. $AB + Bin \cdot (A \oplus B)$
 - D. $A \oplus B \oplus Bin$
108. What is the main operational disadvantage of a standard Parallel Ripple Carry Adder?
- A. It uses too many XOR gates.
 - B. It cannot add negative numbers.
 - C. **The propagation delay accumulates because each stage must wait for the carry from the previous stage.** ✓
 - D. It consumes too much power.
109. Which specific circuit overcomes the delay of a Ripple Carry Adder by predicting the carry bits based solely on the input variables?
- A. Multiplexer
 - B. **Look-Ahead Carry Adder** ✓
 - C. Half Subtractor
 - D. Parity Generator
110. In a Look-Ahead Carry Generator, the "Carry Generate" (G_i) function is mathematically defined as:

- A. $A_i \cdot B_i$ ✓
B. $A_i \oplus B_i$
C. $A_i + B_i$
D. $A'_i \cdot B_i$
111. In a Look-Ahead Carry Generator, the "Carry Propagate" (P_i) function is mathematically defined as:
A. $A_i \cdot B_i$
B. $A_i \oplus B_i$ ✓
C. $A_i + B_i$
D. $A'_i \cdot B_i$
112. Which combinational circuit is known as a "Data Selector" because it routes one of many data inputs to a single output line based on select lines?
A. Decoder
B. Encoder
C. **Multiplexer (MUX)** ✓
D. Demultiplexer
113. A Multiplexer (MUX) is logically referred to as a:
A. Serial-to-Parallel Converter
B. **Parallel-to-Serial Converter** ✓
C. Analog-to-Digital Converter
D. Voltage Divider
114. A MUX with 8 data input lines requires exactly how many Select (Control) lines?
A. 2
B. **3 (Since $2^3 = 8$)** ✓
C. 4
D. 8
115. Aside from routing data, a Multiplexer is extremely useful for directly implementing:
A. Power division.
B. **Any complex Boolean logic function directly without using separate basic gates.** ✓
C. Analog amplification.
D. Frequency modulation.
116. What is the core function of a Demultiplexer (DeMUX)?
A. To combine multiple inputs into one output.
B. **To route a single input signal to one of several output lines based on select lines (Data Distributor).** ✓
C. To add binary numbers.

D. To store data permanently.

117. A Demultiplexer performs the exact same mathematical logic function as a:

- A. Multiplexer
- B. **Decoder (with the data input acting as an Enable line) ✓**
- C. Adder
- D. Flip-flop

118. A Decoder with n input lines will have a maximum of how many output lines?

- A. n
- B. n^2
- C. **2^n ✓**
- D. $2n$

119. If a 1-of-16 decoder with active-LOW outputs exhibits a LOW on the decimal 12 output, what are the binary inputs?

- A. 1010
- B. 0110
- C. **1100 ✓**
- D. 0100

120. A BCD-to-Decimal decoder is alternatively and commonly known as a:

- A. **4-line-to-10-line decoder ✓**
- B. 3-to-8 decoder
- C. priority encoder
- D. multiplexer

121. A BCD-to-7-Segment Decoder (like the 74LS47) accepts a BCD input and drives a 7-segment display. How many input and output lines does it typically have?

- A. 8 inputs, 3 outputs.
- B. **4 inputs, 7 outputs. ✓**
- C. 10 inputs, 7 outputs.
- D. 4 inputs, 10 outputs.

122. If a BCD-to-7 segment decoder receives 0100 on its inputs, which segments become active to display the number?

- A. a, c, f, g
- B. **b, c, f, g (To display the digit '4') ✓**
- C. b, c, e, f
- D. b, d, e, g

123. An 8-to-3 Encoder performs the exact reverse mathematical operation of a:

- A. Multiplexer

- B. **3-to-8 Decoder** ✓
- C. Flip-Flop
- D. Shift Register

124. In a Priority Encoder, if two inputs are activated simultaneously, what happens?

- A. The output is scrambled.
- B. **It outputs the code of the highest-value active input.** ✓
- C. It outputs the code of the lowest-value active input.
- D. The chip resets.

125. If an octal-to-binary priority encoder has its 0, 2, 5, and 6 inputs active simultaneously, the output is:

- A. 101
- B. **110 (Binary for 6, the highest active input)** ✓
- C. 010
- D. 000

126. A combinational circuit that evaluates whether two n-bit numbers are equal, greater, or less than each other is called a:

- A. Adder
- B. Subtractor
- C. **Magnitude Comparator (e.g., 74HC85)** ✓
- D. Parity Generator

127. If a 74HC85 magnitude comparator has $A = 1000$ and $B = 1010$, what are the resulting outputs?

- A. $A > B = 1, A < B = 0, A = B = 0$
- B. **$A > B = 0, A < B = 1, A = B = 0$** ✓
- C. $A > B = 0, A < B = 0, A = B = 1$
- D. $A > B = 1, A < B = 1, A = B = 0$

Chapter 5: Sequential Logic (Flip-Flops, Latches & Timers)

128. Unlike Combinational circuits, Sequential Logic Circuits MUST contain which crucial component to function properly?
- A. Multiplexers
 - B. **Memory elements (feedback loops) to store past states. ✓**
 - C. Only Universal Gates
 - D. Analog components
129. Sequential circuits whose behavior depends heavily on the sequence of past inputs are technically defined as:
- A. **Finite State Machines (FSM) ✓**
 - B. Digital Processors
 - C. Boolean Maps
 - D. Arithmetic Logic Units
130. In FSM theory, a machine where the output depends ONLY on the current internal state is called a:
- A. Mealy Machine
 - B. **Moore Machine ✓**
 - C. Turing Machine
 - D. State Diagram
131. In FSM theory, a machine where the output depends on BOTH the current internal state AND the present external inputs is called a:
- A. **Mealy Machine ✓**
 - B. Moore Machine
 - C. Turing Machine
 - D. Shift Register
132. If a digital system strictly operates using clock edges rather than levels, it is officially referred to as:
- A. Asynchronous
 - B. **Synchronous ✓**
 - C. Combinational
 - D. Analog
133. What is the fundamental difference between a Latch and a Flip-Flop?
- A. Latches require more power.
 - B. **Latches are level-sensitive (transparent), while Flip-Flops are edge-triggered by a clock. ✓**

- C. Latches cannot store data.
 - D. Flip-flops are combinational.
134. In Sequential circuits, the minimum time the data must be stable **before** the active clock edge arrives is called:
- A. **Setup Time (t_{su})** ✓
 - B. Hold Time (t_h)
 - C. Propagation Delay
 - D. Cycle Time
135. The minimum time the data must remain stable **after** the active clock edge has passed is called:
- A. Setup Time (t_{su})
 - B. **Hold Time (t_h)** ✓
 - C. Propagation Delay
 - D. Clock Skew
136. If the Setup or Hold time requirements of a flip-flop are violated, the output may enter an unpredictable, oscillating state known as:
- A. Race condition
 - B. **Metastability** ✓
 - C. Overdrive
 - D. Toggle mode
137. A basic active-HIGH SR latch is typically constructed by cross-coupling which type of logic gates?
- A. AND Gates
 - B. **NOR Gates** ✓
 - C. NAND Gates
 - D. XOR Gates
138. A basic active-LOW SR latch is constructed by cross-coupling which type of logic gates?
- A. AND Gates
 - B. NOR Gates
 - C. **NAND Gates** ✓
 - D. XOR Gates
139. What is the mathematical Characteristic Equation for an SR Flip-Flop?
- A. $Q(t+1) = J \cdot Q'(t) + K' \cdot Q(t)$
 - B. $Q(t+1) = D$
 - C. $Q(t+1) = S + R' \cdot Q(t)$ ✓
 - D. $Q(t+1) = T \oplus Q(t)$

140. What is the mathematical Characteristic Equation for a JK Flip-Flop?

- A. $Q(t+1) = J \cdot Q'(t) + K' \cdot Q(t)$ ✓
- B. $Q(t+1) = D$
- C. $Q(t+1) = S + R' \cdot Q(t)$
- D. $Q(t+1) = T \oplus Q(t)$

141. A flip-flop is formally considered to be in the "SET" state when its outputs are:

- A. $Q = 0, Q' = 1$
- B. $Q = 1, Q' = 0$ ✓
- C. $J = 0, K = 0$
- D. Both outputs are equal.

142. A flip-flop is formally considered to be in the "RESET" state when its outputs are:

- A. $Q = 0, Q' = 1$ ✓
- B. $Q = 1, Q' = 0$
- C. $J = 1, K = 1$
- D. $S = 1, R = 1$

143. For what combinations of inputs will a gated D Latch actually change state?

- A. $D = LOW, EN = LOW$
- B. **When EN = HIGH, the latch is transparent and output follows D** ✓
- C. $D = HIGH, EN = LOW$
- D. It only changes on a clock edge.

144. For an edge-triggered D flip-flop, the output behaves how?

- A. **A change in state can only occur at the active clock edge, taking the value of D.** ✓
- B. The output follows the input continuously at all times.
- C. It toggles on every edge.
- D. It requires J and K inputs to operate.

145. To rapidly create a D Flip-Flop using a JK Flip-Flop, you must:

- A. **Connect J to the data input, and K to the inverted data input (via a NOT gate).** ✓
- B. Tie J and K together directly.
- C. Ground J and set K to 1.
- D. Connect both J and K to the Clock.

146. If you connect the output Q' of a D flip-flop directly back into its own D input, what behavior will the circuit exhibit on every clock pulse?

- A. It will permanently output 1.
- B. It will permanently output 0.
- C. **It will act as a Toggle (divide-by-2 counter).** ✓

- D. It will overheat.
147. A major feature that mathematically distinguishes the J-K flip-flop from the standard D flip-flop is the presence of the:
- A. Clock pulse
 - B. **Toggle condition** ✓
 - C. Clear input
 - D. Preset input
148. In a J-K flip-flop, what occurs when $J = 0$, $K = 0$ and a clock pulse triggers?
- A. The output toggles.
 - B. The output SETs to 1.
 - C. The output RESETs to 0.
 - D. **The output holds its previous state (No Change).** ✓
149. What specific input combination forces a J-K flip-flop into the "Toggle" mode?
- A. $J = 0, K = 0$
 - B. $J = 1, K = 0$
 - C. $J = 0, K = 1$
 - D. **$J = 1, K = 1$** ✓
150. If a J-K flip-flop is continuously held in toggle mode ($J = 1, K = 1$) and fed a 10 kHz clock input, what will be the frequency of the Q output?
- A. 10 kHz
 - B. 20 kHz
 - C. **5 kHz (The toggle mode perfectly divides the clock frequency by 2)** ✓
 - D. Constantly HIGH
151. If you want to make the true output (Q) of a flip-flop zero instantly, entirely independent of the clock cycle, what must you do?
- A. Apply a 0 to the D input.
 - B. **Activate the asynchronous CLEAR (or Reset) pin.** ✓
 - C. Apply 0 to J and 1 to K.
 - D. Turn off Vcc.
152. To uniquely solve the infamous "Race-Around Condition" in older level-triggered JK flip-flops, engineers heavily utilized which configuration?
- A. Adding a D-Latch
 - B. Grounding the J input
 - C. **The Master-Slave Configuration** ✓
 - D. Removing the clock
153. A "One-Shot" timer circuit is technically a type of:

- A. Astable Multivibrator
 - B. **Monostable Multivibrator ✓**
 - C. Bistable Multivibrator
 - D. Decoder
154. Which circuit has ZERO stable states and continuously alternates between high and low, acting as a free-running oscillator clock?
- A. Monostable Multivibrator
 - B. Bistable Multivibrator
 - C. **Astable Multivibrator ✓**
 - D. D-Latch
155. The output pulse width of a nonretriggerable one-shot timer depends primarily on:
- A. The trigger interval
 - B. **An internal Resistor and Capacitor (RC) combination. ✓**
 - C. The clock frequency
 - D. The supply voltage
156. In a standard 555 Timer IC configured as a Monostable multivibrator, what is the formula for the output pulse width (t_W)?
- A. $t_W = R \cdot C$
 - B. $t_W = 1.1 \cdot R \cdot C$ ✓
 - C. $t_W = 1.44 / (R \cdot C)$
 - D. $t_W = 0.693 \cdot R \cdot C$
157. In a 555 Timer configured in standard Astable mode, the frequency of oscillation (f) is determined by the formula:
- A. $f = 1.44 / ((R_1 + 2R_2) \cdot C)$ ✓
 - B. $f = 1.1 \cdot R \cdot C$
 - C. $f = 1 / (R \cdot C)$
 - D. $f = (R_1 + R_2) / C$
158. In a standard 555 Timer IC, Pin 2 is officially known as the:
- A. **Trigger (activates below 1/3 Vcc) ✓**
 - B. Threshold
 - C. Output
 - D. Reset
159. In a standard 555 Timer IC, Pin 6 is officially known as the:
- A. **Threshold (triggers above 2/3 Vcc) ✓**
 - B. Trigger
 - C. Discharge
 - D. Control Voltage

Chapter 6: Registers, Counters and Memory

- 160.** Which sequential circuit type is primarily used to temporarily store and manipulate/shift binary data across systems?
- A. **Shift Register** ✓
 - B. Counter
 - C. Multiplexer
 - D. Comparator
- 161.** To shift an 8-bit word serially (one bit at a time) completely into a standard Shift Register, how many clock pulses are required?
- A. 1
 - B. 4
 - C. **8** ✓
 - D. 16
- 162.** A shift register configuration where data is loaded all at once simultaneously, and then shifted out one bit at a time is called:
- A. SISO
 - B. **PISO (Parallel-In, Serial-Out)** ✓
 - C. SIPO
 - D. PIPO
- 163.** A Shift Register that accepts data one bit at a time sequentially, and outputs all data lines simultaneously is classified as:
- A. SISO
 - B. PIPO
 - C. **SIPO (Serial-In, Parallel-Out)** ✓
 - D. PISO
- 164.** Which term correctly describes a counter where the clock input is connected ONLY to the first flip-flop, and subsequent flip-flops are triggered by the outputs of preceding ones?
- A. Synchronous Counter
 - B. **Asynchronous (Ripple) Counter** ✓
 - C. Ring Counter
 - D. Decade Counter
- 165.** What is a major disadvantage of an Asynchronous (Ripple) Counter operating at very high frequencies?
- A. It uses too much power.
 - B. **Accumulated Propagation Delay (Glitches) because flip-flops do not change state simultaneously.** ✓

- C. It can only count down.
D. It requires a 555 timer to operate.
- 166.** What is the main advantage of a Synchronous Counter over an Asynchronous Counter?
- A. It uses fewer flip-flops.
B. **It eliminates propagation delay ripple, allowing for much higher operating frequencies without decoding glitches. ✓**
C. It doesn't require a clock.
D. It acts as an Analog-to-Digital converter.
- 167.** What is the total Modulo (number of states) of a standard full N -bit binary counter?
- A. N
B. $2N$
C. 2^N ✓
D. N^2
- 168.** A Decade Counter is also officially referred to in literature as a:
- A. **MOD-10 Counter (Counts 0 to 9) ✓**
B. MOD-16 Counter
C. Ring Counter
D. BCD Decoder
- 169.** How do engineers typically force a standard 4-bit binary counter (MOD-16) to act as a MOD-10 (Decade) counter?
- A. By removing 2 flip-flops.
B. By changing the clock frequency to 10 Hz.
C. **By using a NAND gate connected to the asynchronous CLEAR pins to reset the counter when it hits the state 1010 (Decimal 10). ✓**
D. By adding a MUX.
- 170.** If you need to build a counter that counts up to exactly 60 (MOD-60) for a digital clock, what is the absolute minimum number of Flip-Flops required?
- A. 4
B. 5
C. **6 (Since $2^6 = 64$, which safely covers 60 states) ✓**
D. 8
- 171.** If a 1 MHz clock signal is applied to the input of a MOD-16 counter, what is the exact frequency of the output from the final flip-flop?
- A. 1 MHz
B. 500 kHz
C. 125 kHz
D. **62.5 kHz ($1,000,000/16 = 62,500$ Hz) ✓**

172. Which specific digital circuit is formed physically by cascading D-type flip-flops in a closed continuous loop (where the Q output of the last flip-flop feeds directly back to the D input of the first)?
- A. Ripple Counter
 - B. **Ring Counter** ✓
 - C. Multiplexer
 - D. Shift-Right Register
173. A "Twisted Ring Counter" where the inverted output (Q') of the last flip-flop feeds into the input of the first flip-flop is better known as a:
- A. Ripple Counter
 - B. **Johnson Counter** ✓
 - C. BCD Counter
 - D. Program Counter
174. In a Johnson Counter with N flip-flops, how many valid states exist?
- A. N states
 - B. 2^N states
 - C. **$2N$ states (A 4-bit Johnson Counter has 8 states)** ✓
 - D. N^2 states
175. Which type of memory is highly volatile and loses all stored data the moment power is turned off?
- A. **RAM (Random Access Memory)** ✓
 - B. ROM
 - C. Flash Memory
 - D. EPROM
176. SRAM (Static RAM) is extremely fast and used for CPU Cache. What is it primarily built using?
- A. **Flip-Flops** ✓
 - B. Capacitors
 - C. Inductors
 - D. Diodes
177. DRAM (Dynamic RAM) is used for main system memory. Why must it be continuously refreshed?
- A. Because the clock runs too fast.
 - B. **Because it stores data using capacitors, which slowly leak charge over time.** ✓
 - C. Because flip-flops overheat.
 - D. Because it is non-volatile.
178. Which special-purpose CPU register holds the memory address of the NEXT instruction to be executed?

- A. **PC (Program Counter) ✓**
- B. IR (Instruction Register)
- C. MAR (Memory Address Register)
- D. ALU

179. An EPROM (Erasable Programmable Read-Only Memory) chip is uniquely erased by exposing it to:

- A. High voltage
- B. **Ultraviolet (UV) Light ✓**
- C. Magnetism
- D. Radio waves

Chapter 7: DLD Lab Practical and Hardware Components

- 180.** In the DLD hardware lab, which reusable platform is used to physically build and test electronic circuits rapidly without any soldering?
- A. PCB (Printed Circuit Board)
 - B. **Breadboard** ✓
 - C. Multimeter
 - D. Oscilloscope
- 181.** Breadboards have interconnected holes underneath. The horizontal rails at the top and bottom (usually marked red and blue/black) are strictly used for:
- A. Logic gates.
 - B. **Power (Vcc) and Ground connections.** ✓
 - C. Connecting LEDs.
 - D. Bypassing capacitors.
- 182.** The Digital Multimeter (DMM) is a versatile lab instrument utilized primarily to measure:
- A. Light intensity.
 - B. **Voltage, Current, and Resistance.** ✓
 - C. Clock frequencies only.
 - D. Binary code conversions.
- 183.** When using a Multimeter in "Continuity Mode", what does a loud beep indicate?
- A. High voltage is present.
 - B. **There is a direct electrical path (very low resistance / short circuit) between the two probes.** ✓
 - C. The circuit is open/broken.
 - D. The battery is dead.
- 184.** When measuring Current (Amperes) using a Multimeter, how must the probes be connected?
- A. **In series (physically breaking the circuit path).** ✓
 - B. In parallel.
 - C. With power disconnected.
 - D. Across the power supply only.
- 185.** Which logic family typically uses $+5V$ for logic HIGH and $0V$ for logic LOW, and is built using Bipolar Junction Transistors (BJTs)?
- A. CMOS
 - B. **TTL (Transistor-Transistor Logic)** ✓
 - C. ECL

D. Relay Logic

186. What is a major advantage of the CMOS logic family over the older TTL family?

- A. They can withstand thousands of volts safely.
- B. **They consume significantly less static power. ✓**
- C. They use more current.
- D. They don't use binary logic.

187. If an input pin on a standard TTL logic gate is left "floating" (not connected to anything), the IC will inherently interpret it as:

- A. **Logic HIGH (1) ✓**
- B. Logic LOW (0)
- C. An error state
- D. Ground

188. Certain ICs have "Open-Collector" outputs. To make these outputs function properly and pull the logic HIGH, you must externally attach a:

- A. Capacitor
- B. **Pull-up Resistor ✓**
- C. Diode
- D. Inductor

189. Tri-state logic devices feature a unique third output state beyond HIGH and LOW. What is it?

- A. Floating Ground
- B. **High-Impedance (High-Z), effectively disconnecting the output from the bus. ✓**
- C. Metastable State
- D. Mid-Voltage

190. What is the exact industry-standard IC number for a "Quad 2-input NAND Gate"?

- A. **IC 7400 ✓**
- B. IC 7404
- C. IC 7410
- D. IC 7432

191. What is the exact industry-standard IC number for a "Quad 2-input NOR Gate"?

- A. **IC 7402 ✓**
- B. IC 7404
- C. IC 7408
- D. IC 7486

192. What is the exact industry-standard IC number for a "Hex Inverter" (NOT Gate)?

- A. IC 7400

- B. **IC 7404** ✓
- C. IC 7410
- D. IC 7408

193. What is the exact industry-standard IC number for a "Quad 2-input AND Gate"?

- A. IC 7400
- B. IC 7404
- C. **IC 7408** ✓
- D. IC 7432

194. What is the exact industry-standard IC number for a "Triple 3-input AND Gate"?

- A. **IC 7411** ✓
- B. IC 7410
- C. IC 7408
- D. IC 7404

195. What is the exact industry-standard IC number for a "Quad 2-input OR Gate"?

- A. IC 7400
- B. IC 7404
- C. IC 7408
- D. **IC 7432** ✓

196. What is the exact industry-standard IC number for a "Quad 2-input XOR Gate"?

- A. IC 7400
- B. IC 7404
- C. **IC 7486** ✓
- D. IC 7410

197. What is the exact industry-standard IC number for a "Triple 3-input NAND Gate"?

- A. IC 7400
- B. IC 7404
- C. **IC 7410** ✓
- D. IC 7486

198. What is the exact industry-standard IC number for a "Quad 2-input EX-NOR Gate"?

- A. **IC 74266** ✓
- B. IC 7410
- C. IC 7427
- D. IC 4075

199. In a standard 14-pin IC diagram, the pin designated as "GND" stands for:

- A. Gate Node

- B. **Ground (0V Reference, usually Pin 7) ✓**
 - C. Generated Output
 - D. General Data
200. When wiring an IC 7404 (Hex Inverter) on a breadboard, Pin 14 is generally connected to:
- A. Ground (GND)
 - B. **Power Supply (+5V Vcc) ✓**
 - C. The input signal.
 - D. The LED anode.
201. When an LED is connected to a 5V TTL circuit on a breadboard, what MUST be placed in series with it to limit current and prevent it from burning out?
- A. A Capacitor
 - B. **A Resistor (typically 220 to 330 Ohms) ✓**
 - C. An Inductor
 - D. A Diode
202. The mathematical formula used to perfectly calculate the exact value of the current-limiting resistor required for an LED in a circuit is:
- A. **$R = (V_{source} - V_{LED}) / I_{LED}$ ✓**
 - B. $R = V_{source} \times I_{LED}$
 - C. $R = V_{LED} / V_{source}$
 - D. $R = I_{LED} / V_{source}$
203. In a "Common Anode" 7-segment display, all the positive terminals (anodes) of the internal LEDs are tied together to Vcc. To light up a specific segment, you must apply a:
- A. Logic HIGH (1)
 - B. **Logic LOW (0 / Ground) ✓**
 - C. High-Z state
 - D. Tri-state pulse
204. In a "Common Cathode" 7-segment display, all the negative terminals (cathodes) are tied to Ground. To light up a specific segment, you must apply a:
- A. **Logic HIGH (1 / Vcc) ✓**
 - B. Logic LOW (0)
 - C. High-Z state
 - D. Tri-state pulse

Chapter 8: Advanced Mixed MCQs & Architecture

205. A logic circuit with an output $X = \overline{AB} + \overline{ABC}$ consists of which minimum set of gates?
- A. Two AND gates and one OR gate
 - B. **Two AND gates, one OR gate, and two inverters ✓**
 - C. Two AND gates, two OR gates, and two inverters
 - D. Two AND gates, one OR gate, and three inverters
206. Which logic family requires the least power, making it ideal for battery-operated devices like laptops and smartphones?
- A. TTL
 - B. ECL
 - C. **CMOS ✓**
 - D. DTL
207. To measure the period of a fast pulse waveform accurately, you must use a/an:
- A. Digital Multimeter
 - B. Logic Probe
 - C. **Oscilloscope ✓**
 - D. Logic Pulser
208. Once you physically measure the period (T) of a pulse waveform on an oscilloscope, the frequency is found by:
- A. Measuring the duty cycle
 - B. **Finding the reciprocal of the period ($f = 1/T$) ✓**
 - C. Multiplying by the amplitude
 - D. Adjusting the V/div knob
209. What does PLD stand for in digital electronics?
- A. Primary Logic Decoder
 - B. **Programmable Logic Device ✓**
 - C. Printed Logic Design
 - D. Process Level Diagram
210. Which type of PLD contains both a programmable AND array and a programmable OR array?
- A. PAL (Programmable Array Logic)
 - B. **PLA (Programmable Logic Array) ✓**
 - C. GAL (Generic Array Logic)
 - D. PROM

211. Which type of PLD has a programmable AND array but a strictly FIXED OR array?
- A. PLA
 - B. **PAL (Programmable Array Logic) ✓**
 - C. CPLD
 - D. FPGA
212. In a Mealy Machine (FSM), the output depends exclusively on:
- A. Only the current state.
 - B. **Both the current state and the present external inputs. ✓**
 - C. Only the next state.
 - D. The clock frequency.
213. In a Moore Machine (FSM), the output depends exclusively on:
- A. **Only the current state. ✓**
 - B. Both the current state and present inputs.
 - C. The memory address register.
 - D. The arithmetic logic unit.
214. Which flip-flop is heavily used in building Shift Registers because of its ability to transfer exact data on a clock edge?
- A. SR Flip-Flop
 - B. JK Flip-Flop
 - C. **D Flip-Flop (Data) ✓**
 - D. T Flip-Flop
215. Which flip-flop is heavily used in building Binary Counters because of its ability to divide frequencies by 2?
- A. SR Flip-Flop
 - B. D Flip-Flop
 - C. **T Flip-Flop (Toggle) ✓**
 - D. Gated Latch
216. The phenomenon where the output of a logic gate flips unpredictably multiple times during a single clock pulse (found in level-triggered SR/JK flip flops) is known as:
- A. Propagation Delay
 - B. **The Race-Around Condition ✓**
 - C. Clock Skew
 - D. Setup Time Violation
217. A 4-bit binary synchronous counter will require exactly how many flip-flops to build?
- A. 2
 - B. **4 ✓**

- C. 8
D. 16
218. If you need to count up to the number 250 in binary, what is the absolute minimum number of flip-flops required for the counter?
- A. 4
B. 6
C. **8 (Since $2^8 = 256$) ✓**
D. 10
219. An EEPROM memory chip is unique because it can be erased:
- A. Using Ultraviolet light.
B. By removing the battery.
C. **Electrically, byte by byte, while still in the circuit. ✓**
D. Only once at the factory.
220. Flash Memory (used in USB drives and SSDs) is technically a specialized high-density type of:
- A. SRAM
B. DRAM
C. **EEPROM (Erased in blocks rather than bytes) ✓**
D. ROM
221. The "Flags Register" in a microprocessor is used primarily to:
- A. Store the next memory address.
B. **Store boolean conditions (Carry, Zero, Overflow) based on the last ALU operation. ✓**
C. Count the number of clock pulses.
D. Route data to the printer.
222. Which special-purpose register (SPR) holds the actual data that was just fetched from memory, or is waiting to be written to memory?
- A. MAR (Memory Address Register)
B. PC (Program Counter)
C. **MDR (Memory Data Register) ✓**
D. IR (Instruction Register)
223. If a single D flip-flop divides the input clock frequency by 2, what will be the output frequency if you cascade three D flip-flops in a ripple counter and input a 100 kHz clock?
- A. 50 kHz
B. 25 kHz
C. **12.5 kHz (Divided by $2^3 = 8$) ✓**
D. 100 kHz

224. The boolean expression $(A + B + C)(A + B + C)$ simplifies dramatically to:

- A. $A + B + C$ (Idempotent Law: $X \cdot X = X$) ✓
- B. 1
- C. 0
- D. $A \cdot B \cdot C$

***** BEST OF LUCK FOR YOUR FINAL EXAMS! *****

You have mastered exactly 224 highly targeted, in-depth MCQs for SE-455 DLD. You are fully equipped to secure a perfect 100% in your Objective Paper.

A Message from the Author

Assalam-o-Alaikum! I am Shaikh Mahad, a student of UBIT (2025 Batch).

I am currently in my third semester (Year 2026) learning this course. When I started my preparation, I couldn't find proper consolidated materials to practice from. Realizing the gap, I took the initiative to structure, design, and compile this comprehensive PDF initially for my own studies. Knowing how helpful this could be to others, I massively expanded it into this Ultimate MCQ Bank to help all present and future students master Digital Logic Design with ease. Please note that this document was structured and enhanced with the help of AI. My goal is that this document serves as a valuable tool in the academic journey of everyone learning this course now, and in the future batches to come.

Let's Connect & Collaborate:

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